

Tips and techniques for industrial job search and interviews

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12/05/20

Outline

- We will present informal tips and techniques for job search for typical industrial jobs
- Changes in global environment for SW professionals will strongly influence jobs search, and this will also be covered
- Based on speaker's 20 year experience in mostly hiring, and some interviewing
- Be sure to attend professional presentations or read books about it if you are more interested
- We recently added information on your electronic portfolio
- Appendix I: how to update your resume after taking CSC 648-848 Software Engineering class

Example advertisement – check underlined segments

- **Sr. Software Engineer**

- **Job Description**

The Software Engineer will work as part of an agile multi-disciplinary team to develop the software components of an enterprise-scale hospital information system. The individual must be a team-player and willing to function as a designer, developer, tester, and an analyst as required to achieve the goals of the team.

- **Specific Responsibilities:**

- 5+ years of professional experience developing commercial or enterprise-scale software products
- 3+ years of development experience with Java and J2EE (EJB, Servlets, and JSP)
- XP, Agile development experience is preferred
- Healthcare domain knowledge is preferred
- Exposure to multiple DBMS systems is preferred
- Understands concepts of the software development lifecycle
- Ability to function as a designer, developer, tester and to some degree, an analyst
- Must possess strong organization and communication skills
- Must exhibit a sense of and demonstrate responsibility, focus on delivery, and ability to work independently with appropriate technical direction
- Comfortable in a fast-paced, team-oriented environment
- Strong written and verbal skills from both business and technical perspectives

Environment

- Computer Engineering – one of the rare industries doing well and hiring
 - Demand strong, pays well
 - New areas as of late: mobile, cloud, social networking, big data, AI/ML
 - Skills in addition to core technical skills are also needed:
 - Knowledge of SE lifecycle
 - Being able to provide great User experience a good skill
 - Global SW development
 - Knowledge of foreign cultures and languages helps a lot
 - Communication (oral and writing)
- ➔ **Combination of skills is needed – combination of technical, organizational and communicational skills**

New Skills Needed

- Today – you need **additional** skills:
 - Project and SW engineering skills
 - Ability to participate, design or manage global SW development
 - Understand users and application domains (bio, art, finance)
 - Ability to create systems that delight the users
 - Ability to architect and deliver systems from components done by others
 - Interpersonal and communication skills

Don't forget: knowing basics is still necessary (data structures, programming, architecture)

Finding a job

- Network of friends and professional associates is the key – most important!

Also

- Social networking sites
- Company WWW sites
- On-line search services
- Recruiters
- Job Fairs
- Newspapers

Try it all....

Job related links on our WWW site

- Student employment page (listings, SFSU services, CS resources, SFSU recruiting events, slides from this talk
 - <http://cs.sfsu.edu/employment>
- SFSU career center (events, resume prep, mockup interviews)
 - <http://www.sfsu.edu/%7Ecareer/>

Finding a job - Recruiters

- Understand how they operate
- Ask them who are their clients and past successes
- Note that many companies do not like external recruiters and use their own
- Be clear on what companies you already contacted
- Are they exclusive, what are your obligations
- Read all documents before you sign
- Do not give them your resume before you know the rules, they might flood the WWW with it
- Check how and what they charge

Finding a job – on-line or off-line application submission

- **Goal:** to pass the first filter (HR or some kind of KW screening program)
- Salary: leave it open, do not enter numbers
- Be flexible, agree on travel etc.
- Cover letter or cover e-mail
- Resume
- Many times sending surface mail gets better attention, especially if you know somebody.
- Lately, on-line submissions had little value, not even a response is given
- Try to submit some kind of cover letter or e-mail

Cover letter or introductory e-mail

1. What are you applying for
2. How did you find about this (advertisement, your own initiative, reference from somebody)
3. Summary of your skills and qualifications of specific interest to this company and opening (here you customize it for this job!)
4. Close by saying that your CV is attached and offering and suggesting some kind of follow up

Total size up to half to three quarters of a page max.

No typos, good style and proper English (have your friends check it)

Resume

1. Objective (optional)
 2. Skills (be honest)
 3. Awards, something very special
 4. Education AND experience (whichever is more impressive goes first)
 5. Hobbies etc.
 6. Visa status (if applicable)
 7. References: available upon request (Do not give them before hand. Have your set of references ready and send them your resume).
- 1 page!!!! (unless you are very senior or going for academia)

Resume >> Objective (optional)

- BUT this can be optional so you do not have to customize it for each job – address objectives in the cover letter
- Make it not too general, not too specific. Avoid useless stuff (“to find challenging position”). At least mention whether it is in R&D, or sales support, QA etc (I.e. what area of work)
- Suggestion: cover position type (internship or permanent or management) and mention area e.g. development, QA, support. Not much else

Resume >> Skills

Group it rather than provide long single list

- Technical skills (OS, languages, tools)
- Communication skills
- Organizational skills
- Languages spoke, knowledge of other cultures

Resume >> Experience

- Need to clearly denote company, position, time, with nice layout
- Speak in past tense, with verbs like “developed, designed, delivered, implemented...”
- Explain WHAT was done and what YOU did. Say also WHAT WAS UNIQUE about it (fastest, most reliable, done in short time, sold well...)
- Avoid “laundry list” (I.e. flat list of items)
- Did you supervise somebody, did you talk to customers outside of your company? Did you collaborate with sales, marketing?
- Give reference to product names, WWW sites, Github link

Resume > Experience – what if you do not have job experience?

- Briefly explain your projects, MS Thesis, any significant activity where there was a deliverable, work in a team, work with somebody else etc.
- If you worked in some other jobs, where skills are transferable and relevant, say it (supervised somebody in father's mechanic shop etc.). Good organizational skills are transferable: if you managed waiters it means you can deal with people!
- Bring your “portfolio” such as code prints, demos, copies of project documentation (like CSC 640 stuff)

(This is why we have all these project exercises in our classes)

Resume “look and feel”

- Nice formatting and layout
 - Easy to read
 - Sections clearly marked
 - Clean print (no poor copy)
 - Check typos, errors, English
 - If on-line, check formatting and make sure it looks right on the receiving end. Use PDF.
- ➔ Has to look well organized, done with care and professional
- ➔ Check the printed version to make sure it comes OK

Do the homework before any contact with the company

- About the company
- Products
- Management
- Market share
- Competitors
- Download and try products
- Read reviews about the company and products
- Check the people if you can and find out about them
- Ask around
- Use WWW, Google

A word on your social network pages

- Some employers may ask PW on your facebook etc.
- Be careful about what you post on social networking even for private use!!!!
- No privacy in digital world
- Not legal to ask for PW but they still might do
- Some advice
 - <http://mashable.com/2012/04/08/employer-facebook-password/>
 - <http://articles.latimes.com/2012/mar/20/business/la-fi-tn-employers-seek-your-access-to-facebook-20120320>

Prepare your electronic portfolio

- Have a well designed (simple is OK) WWW page where you nicely organize your projects (provide CV too)
- For each project provide a short description and WWW link
- Provide github link to code wherever possible
 - Make sure each folder is named appropriately with useful easy to understand name
 - Provide readme when necessary
 - Have code have header and some inline comments
 - Use proper coding style (variable names, indentation)
 - Add design documents, research papers too

Update your CV/Resume after CSC 648-848 SE class

- See Appendix I

Phone screening

- **Their Goals:** Used to filter candidates before inviting. Saves time and cost for all
- **Your Goal:** impress them, pass their test. You also must to learn about the company and position.
- Done by recruiters, HR or sometimes technical people
- Speak clearly, be articulate.
- Listen, and do not talk too much at the beginning, do not interrupt them.

Phone screening (2)

- Questions asked:
 - Going over the resume and clarifying it
 - Possible fit
 - Technical screening i.e. tests – very basic
 - Asking about your key strengths and skills and possible fit as you see it

Phone screening (3)

- Your favorite projects
- Why are you leaving. Tricky. Be honest if there were problems but stay at high level. Do NOT criticize anybody
- Any questions you have (ask about the job, the company, management etc.)
- Do not ask about money. If asked, stay vague and avoid it. Do not give them any numbers!
- Example:
 - They: What is the salary you want?
 - You: What is the range for my education and experience
 - They: \$ 50 – 70 K
 - You: I believe that my experience qualifies me for upper 60s

Phone screening (4)

- They usually ask you if you have questions. Try this:
 - Company vision
 - Expectations for the position you are interviewing
 - What are the key company competitors and problems that you could solve
 - Opportunity for advancement

Phone Screening (4)

- Visa:
 - Let them ask. If asked, be honest.
 - If they are reluctant, offer to pay costs if they do the paperwork
- Benefits:
 - Medical (what plans, how much you pay)
 - Stock (long story)
 - Vacation
 - Sick days

Interview – you are invited!

Do the homework first

- Check the company, management, products, competitors
- Download and play with their products. Write down comments (questions, suggested improvements)
- Know as much as possible about everybody on the itinerary
- If you are working with recruiter, they should help you.
- Good companies always have an interview host, ask him/her for info
- Refresh on your past work and portfolio
- Think in advance what would you do in next product if you were in that company. How about their competition? Prepare questions.

Interview

>> basics

- Dress neatly and nice. Men: Suit is preferred or at least nice pants, jacket and tie. This holds even if the company CEO is barefoot. Women: professional, not too flashy
- Show up on time
- Check exact directions and office numbers before. Bring their phone number with you
- Bring itinerary
- Bring resume (several copies) and your portfolio (projects, papers).
- Eat before.

Interview >> first and other non-technical impressions - IMPORTANT

- Good eye contact and handshake
- Manner of dress
- Speak clearly and loud enough
- Be happy, smile. Be confident, not tense, not too relaxed
- Sitting: not laid back, not too much forward.
- Be a good listener, do not interrupt, especially at the beginning. Use that to collect info and adjust what you will say.
- But, don't be passive either
- Show enthusiasm
- Eat slowly. Order very small meal because they will be asking you questions all the time. No alcohol, no expensive meals.
- No jokes

ALL

VIDEOS

NEWS

IMAGES

MAI

The way you **shake** the interviewer's hand could seal your chances of **employment** either way. Making eye contact and offering a firm **handshake** exhibits confidence and professionalism. ... A **handshake** is appropriate at the end of your **interview**; however, **shaking hands** also is appropriate before and **during** your **interview**.

 Chron.com › work › shake-intervie...

[Do You Shake an Interviewer's Hand After an Interview? | Chron.com](#)

 About Featured Snippets  Feedback

PEOPLE ALSO ASK

Offer up a firm handshake.
When **you** meet the **interviewer**, smile and **shake hands** firmly. ...
Both men and women **should** give a full-palm handshake when they meet a person for the first time. Feb 7, 2017

<https://theundercoverrecruiter.com/5-essential-interview-techniques-you-should-know/>

5 important things for the interview

- <https://theundercoverrecruiter.com/5-essential-interview-techniques-you-should-know/>

Interview >> first and other non-technical impressions (2)

- Be sensitive to cultural issues and common behaviors of different nationalities etc.
- You will be interviewed by people who come from all over the world
- Respect “personal space” of others (keep the distance, don’t hug them etc. – some cultures do not like that)

Interview >> typical questions

- Verifying your entries in the resume. Checking that you really know what you say you know
- Other strengths: communication, organizations;
- Technical tests (good luck)
- Puzzles, to see your reasoning and how you deal under stress
- Tell me about yourself (this means professional stuff only please)
- Your most favorite project
- Did you work with anybody outside of your group/company?

Interview >> typical questions

- How do you deal with a change
- How do you deal with conflict (technical, personal)
- What are your weaknesses (prepare something on this)

Interview >> other discussion

- They will present you their products, management organization and challenges. If not, ask them about this.
- Comment and engage in this part of the interview. Show them that you did your homework
- Ask tough questions – you will get more respect.
- Show enthusiasm. Be active. People do not like that others are not interested in their company
- Use the questions about company/products that you prepared. Let them know that you use the product. Give them feedback. Know who did what in the product when they interview you and ask them focused questions (how do you do this, why did you do this, did you think of that). They love it!

Interview >> what should you ask

- Job expectation
- About the company, product directions
- Company vision
- Company values
- Challenges they face (technical, competition, business)
- Customers
- Management and their style
- Your immediate group and management
- Culture

(OK to bring written questions)

Interview > > compensation

- Do not start this subject, let them do
- HR would probably tell you about benefits up front
- Do not offer salary figure, not even the range, let them suggest (use previous example)
- Ask about the whole package (salary, benefits, vacation, sick days, stock...)
- Ask what do they pay for comparable skills and experience
- Do not start visa questions, do it only if they ask or when you think you are close to getting the offer

Follow up

- If nothing happens in 2 weeks after the interview, call them or e-mail.
- Be gentle, do not push, but reiterate your enthusiasm and fit
- Offer is not official until you have it in writing. Phone does not count (although if they renege after the phone offer, you are better not to work there anyway).

Appendix I

Updating your résumé and portfolio after CSC 648-848 class

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December 2020

Objective

- Here we outline some suggestions how to update your resume and portfolio with what you learned and have done in CSC 648-848
- Be sure to also consult other resources
- Don't forget CS WWW page on employment with
 - Jobs
 - <http://cs.sfsu.edu/employment/external-student>
 - Tips on tips for job search, interviews and resume prep by Prof. D. Petkovic
 - <http://cs.sfsu.edu/employment>
- Include info about doing it all over ZOOM!

About CSC 648-848

- The class is a bit unique in that it gives you close to real life experience in working in a team to develop real SW app – best next thing to real job experience
- We simulate real work environment
- The class covers *full* SW lifecycle
- In addition, some students get exposure to global SE
- Class syllabus for the reference
 - <http://cs.sfsu.edu/courses.html> (look for CSC 648)
- Research paper
 - <http://ieeexplore.ieee.org/document/7757406/?reload=true>

Suggested updates to the resume

Under SKILLS

- Add the SW tools you learned and used in the class (node.js, bootstrap etc.)
- Add other tools you used for project management like github, slack, Trello etc. also ZOOM
- Have a line item on SW Engineering Process and say something like:
 - “Knowledge of full SE lifecycle including best practices of SE teamwork and SE processes such as Agile/SCUM and User Centered Design for providing good user experience”
- Also add something like this as an item: “experience in running/participating in fully on-line SW development”

Resume: Under “selected school projects” (after “education” item)

- It is good to list up to 3 selected school projects after *education* item. The one related to CSC 648-848 might be:
- For general team members:
 - Worked in a team of <5-7> students to develop full WWW application <app name> using modern teamwork, SE processes and tools. My role was <list your role>.
- For team leads:
 - Add that you were leading a team and explain.
- Those working in global teams add something like
 - Team included students from Fulda University, Germany
- Add URL of the app (but make sure you move it from school server since we need to shut them down after the class)
- Provide link to github team repository – make use it has the right permissions OR better yet copy it on your github
- Mention that the team project were developed fully on-line using ZOOM

Your portfolio

- Make a hard-copy of Millstone 5 folder with all the documents and have it ready in a nice folder for the interview. Employers love to browse the design documents
- After the class move the app to your server and make sure it runs OR record the demo on video and have it on our WWW site
- Clean up and organize github
- Provide link to giuthub team repository – make use it has the right permissions OR better yet copy it on your github

GOOD LUCK!